

GLA UNIVERSITY, GREATER NOIDA

Greater Noida – 201310, Uttar Pradesh, India



URBAN SERVICE RESOLUTION

INTELLIGENCE PLATFORM

SOFTWARE REQUIREMENT SPECIFICATION

Version 1.0

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING - ARTIFICIAL INTELLIGENCE & MACHILNE LEARNIG

Submitted By

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SCHOOL / DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING / APPLICATIONS

GLA UNIVERSITY, GREATER NOIDA – 201310

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Revision History

Version	Date	Change	Reason
1.0	04-09-2026	Initial version	First project SRS draft

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the Urban Service Resolution Intelligence Platform v1.0. It defines the data inputs, prediction outputs, model-backed functions, validation requirements, system boundaries, and acceptance criteria against which the implementation will be evaluated.

1.2 Scope

The system shall analyze urban service-request records and provide machine-learning-based estimates of service resolution time using the available request attributes. It shall also support exploratory analysis and selected classification, clustering, and NLP experiments required by the project. The system does not place service requests with agencies, dispatch field workers, or guarantee actual resolution time.

1.3 Definitions

Term	Definition
Resolution Days	Time between Created Date and Closed Date, used as the main regression target.
Service Request	A recorded urban complaint or service request represented by one dataset row.
High-Delay Class	A binary label for the high-delay vs normal-delay classification experiment.
Target Leakage	Using information unavailable at prediction time, such as Closed Date when predicting Resolution Days.
Model Dataset	Cleaned rows for which Resolution Days is available for supervised training.

2. Overall Description

2.1 User Classes

User class	Who they are	What they may do
Project User / Analyst	User analyzing urban service requests.	Load data, inspect quality, view predictions and analytical outputs.
Project Developer	Person maintaining the ML pipeline.	Prepare data, train models, compare models and evaluate results.

2.2 Assumptions and Dependencies

- The urban service-request CSV dataset shall be available locally in the expected format.
- Python and required data-science/ML libraries shall be available in the development environment.
- Supervised training depends on non-missing Resolution Days values.

2.3 Constraints

- C1** The inspected sample contains 10,000 rows and 44 original columns.
- C2** The selected project dataset contains 18 columns, while the current supervised model dataset contains 3,901 rows.
- C3** Closed Date cannot be used as a prediction feature because it is used to calculate Resolution Days.

3. Data Source and Baseline

The currently inspected dataset is an urban service-request CSV file named export.csv. The notebook loaded a 10,000-row sample from the larger dataset. The sample contains 44 columns, of which 18 project-relevant columns were selected. After creating and cleaning Resolution Days, 3,901 rows currently have a usable target for supervised modeling.

Item	Current project value	Required
Dataset name	export.csv — urban service-request dataset	Yes
Source and URL	To be recorded from the original dataset source.	Yes
Licence / terms of use	To be verified from the original dataset source.	Yes
Rows inspected	10,000 sample rows	Yes
Original columns	44	Yes
Selected project columns	18	Yes
Current model rows	3,901	Yes
Prediction target	Resolution Days	Yes
Task type	Predicting a number (regression)	Yes
Class balance	N/A for primary regression	Yes
Primary evaluation metric	MAE	Yes
Why MAE	Expresses absolute prediction error in days.	Yes

Level 0 baseline	Predict training-set mean Resolution Days for every test row.	Yes
Level 0 score	To be calculated after train/test split.	Yes
Known data-quality problems	Missing values; 6,094 missing Closed Date; 9,974 missing Due Date; 5 negative Resolution Days identified and removed.	Yes
Level 1 baseline	Linear Regression	Later

Primary Model Target

Resolution Days is derived from Created Date and Closed Date. Negative values are invalid. Rows without a usable target are excluded from supervised regression. Due Date is retained for analysis but is not assumed to be a primary training feature.

4. System Architecture

The architecture separates ingestion, preparation, training, evaluation and prediction so that target leakage is avoided and models can be compared consistently.

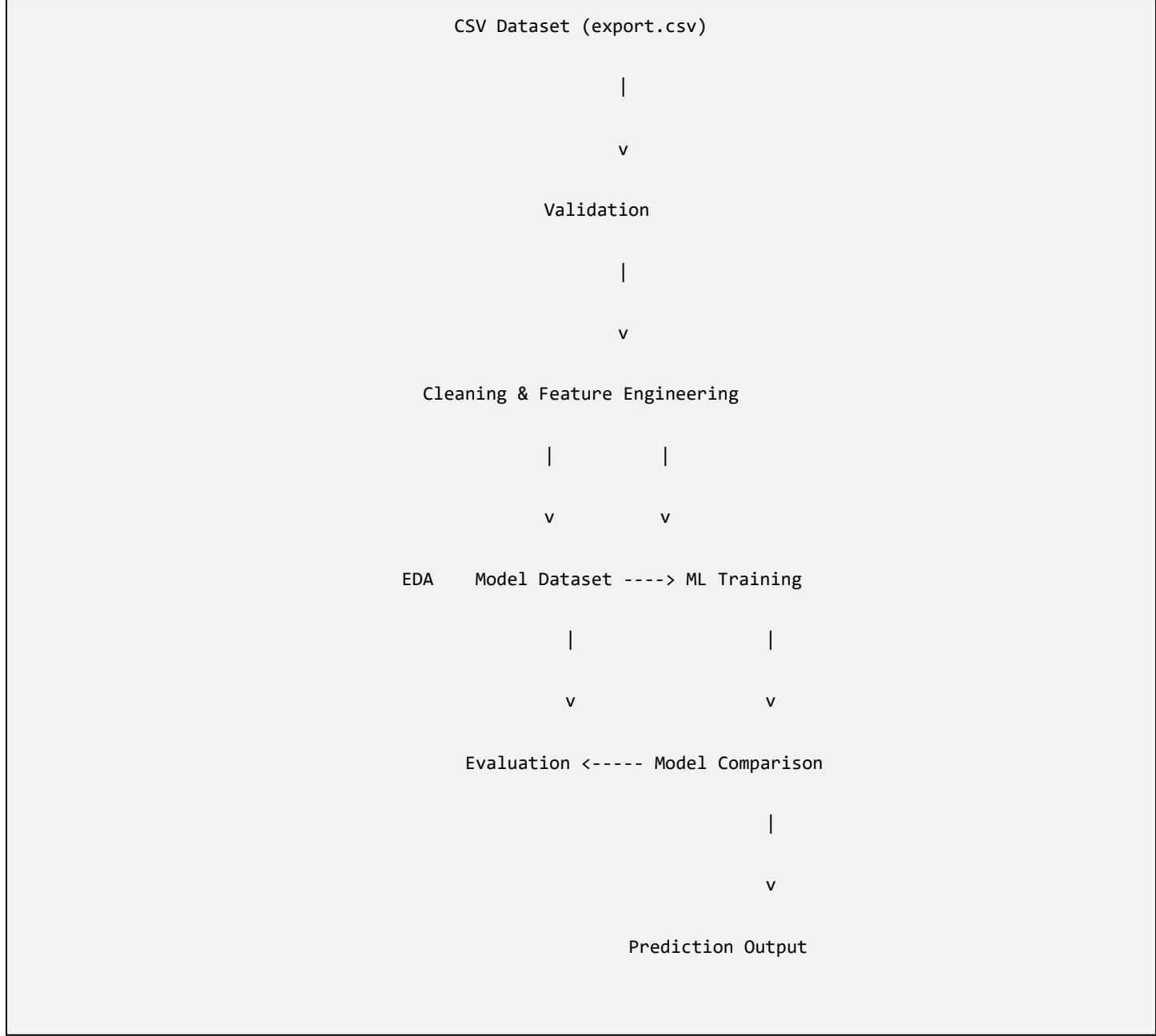


Figure 1: System architecture — data flow from ingestion to prediction output

Component	Single responsibility
Data Ingestion	Load the urban service-request CSV and verify required columns.
Cleaning / Feature Engineering	Handle missing/invalid values and create prediction-ready features.
EDA	Measure distributions, missingness, outliers and service patterns.
ML Training	Train requested regression, classification, NLP and clustering experiments.
Evaluation	Measure model performance with appropriate metrics.
Prediction Output	Return predicted resolution time and/or analytical results.

5. Functional Requirements

5.1 Data Preparation and Analysis

FR-1.1 The system shall load the urban service-request CSV dataset from the configured local path.

FR-1.2 The system shall report whether the configured dataset file exists before loading it.

FR-1.3 The system shall report the number of rows and columns in the loaded sample.

FR-1.4 The system shall retain the project-relevant service-request fields selected for analysis.

FR-1.5 The system shall create Resolution Days from Created Date and Closed Date where both timestamps are usable.

FR-1.6 The system shall remove invalid negative Resolution Days values from the supervised-learning dataset.

FR-1.7 The system shall report missing values for each model-relevant column.

FR-1.8 The system shall exclude rows without a usable Resolution Days target from supervised regression training.

FR-1.9 The system shall display descriptive statistics for Resolution Days before model training.

5.2 Model-backed Requirements

FR-2.1 The system shall accept prediction-time service-request attributes available before resolution.

FR-2.2 The system shall reject prediction input missing a required model feature and identify that feature.

FR-2.3 The system shall return a predicted Resolution Days value for valid input.

FR-2.4 The system shall label the returned prediction as an estimate.

FR-2.5 The system shall provide the Level 0 baseline when the trained model is unavailable.

FR-2.6 The system shall compare trained models using a consistent validation procedure.

FR-2.7 The system shall provide the high-delay vs normal-delay classification output.

FR-2.8 The system shall provide NLP/text-processing output for Problem Detail.

FR-2.9 The system shall provide clustering labels for service-request grouping.

ML Requirements

The project shall use the requested ML topics in meaningful components: linear, multiple and non-linear regression; gradient descent; regularization; logistic regression; PCA; SVD; SVM; decision trees; Naive Bayes; k-means; hierarchical clustering; bagging; boosting; AdaBoost; NLP with spaCy/NLTK; text classification; sentiment analysis; and an introductory CNN experiment. Cross-validation, bias-variance analysis, confusion matrix, and precision-recall shall be used for validation and evaluation.

6. Non-Functional Requirements

ID	Requirement
NFR-1	The prediction component shall return a result within 4 seconds for a single valid request in the local development environment.
NFR-2	The system shall keep target-derived fields separate from prediction inputs to prevent target leakage.
NFR-3	Each model experiment shall record model name, validation score, test score when available, and evaluation metric.
NFR-4	The notebook/application shall organize preparation, training, evaluation and output into reproducible execution steps.

7. Build Plan

7.1 Minimum Viable Product

FR / NFR ID	Requirement	Why in MVP
FR-1.1	Load dataset.	All downstream functions depend on the project data.
FR-1.5	Create Resolution Days.	Defines the main measurable prediction target.
FR-1.6	Remove invalid target values.	Prevents invalid labels from entering training.
FR-2.3	Return Resolution Days prediction.	This is the core ML outcome.
FR-2.5	Level 0 fallback.	Keeps the system demonstrable if the trained model is unavailable.
NFR-2	Prevent target leakage.	Ensures prediction inputs reflect information available at prediction time.
NFR-3	Record model evaluation.	Supports evidence-based model selection.

Deliberately not in the MVP: the complete set of alternative ML, NLP and clustering experiments; these are added after the core pipeline is working.

7.2 Build Order

Stage	Deliverable	IDs	Done when
Sprint I — 27 Sep 2026	Prototype / initial working model	FR-1.1–FR-1.9; FR-2.1–FR-2.5; NFR-2–NFR-3	Data loads, target is clean, baseline is measurable, and first predictions are returned.
Sprint II — 17 Oct 2026	50% functional application	FR-2.6–FR-2.7; NFR-1	Validation and core regression/classification models can be compared.
Sprint III — 7 Nov 2026	75–80% functional application	FR-2.8–FR-2.9; NFR-4	NLP, clustering and remaining ML experiments are integrated.
Testing — 17 Nov 2026	Test report and bug log	FR-1.1–FR-2.9; NFR-1–NFR-4	Functional and evaluation tests are recorded.
Deployment & Documentation — 30 Nov 2026	Deployed application + documentation	All release IDs	Application and documentation are available for demonstration.
Final Presentation — 5–10 Dec 2026	Final report, presentation and viva	All release IDs	Final system and model comparison are demonstrated.

8. Data Requirements

8.1 Entities

Primary entity: Urban Service Request — one urban service-request record used for analysis and ML.

Field	Type	Constraint / Notes
Unique Key	Identifier	Unique record identifier where available.
Created Date	Datetime	Used to derive Resolution Days; available before resolution.
Closed Date	Datetime	Used to derive target; excluded from prediction features.
Agency / Agency Name	Categorical	Agency information.
Problem	Categorical	Problem category.
Problem Detail	Text	Used for NLP/text experiments.
Location Type / City / Borough	Categorical	Location attributes.
Incident Zip / Council District	Numeric/Categorical	Geographic/administrative attributes.
Community Board / Police Precinct	Categorical	Administrative attributes.
Open Data Channel Type	Categorical	Request submission channel.
Latitude / Longitude	Numeric	Geographic coordinates.
Status	Categorical	Service-request status.
Due Date	Datetime	Retained for analysis; primary training use is not assumed.

Resolution Days	Numeric	Main target; non-negative and non-missing for regression training.

8.2 Entity Relationship Diagram

Not required for v1.0. The current implementation is primarily dataset/model focused.

9. Design Decisions and Alternatives Considered

#	Decision	Alternative rejected	Reason
D1	Use Resolution Days as the primary regression target.	Predict Closed Date directly.	Resolution Days is a directly measurable duration derived from the two available timestamps.
D2	Exclude Closed Date from model inputs.	Use every selected column as a feature.	Closed Date is used to calculate the target and would leak future information into prediction inputs.
D3	Use MAE for the primary regression evaluation.	Use classification accuracy for the main task.	The main task predicts a numeric duration, so error in days is directly interpretable.

10. Acceptance Criteria

ID	Scenario	Expected outcome
AC-1	Load export.csv.	File availability is reported and the sample loads with row/column counts.
AC-2	Create Resolution Days from valid timestamps.	Resolution Days is created and negative values are identified as invalid.
AC-3	Run supervised preparation with missing Resolution Days.	Rows without a usable target are excluded and the remaining model-row count is reported.
AC-4	Submit valid prediction input after model training.	The system returns a Resolution Days estimate with its model/evaluation context.

11. Out of Scope for This Release

Automatic agency dispatch. The system will not assign field workers or directly create operational work orders.

Guaranteed real-time resolution. Predictions are estimates and are not guaranteed completion times.

Production-scale cloud infrastructure. Full distributed/cloud deployment, continuous streaming, and production MLOps are outside v1.0.

Appendix A: State Model

Not required for v1.0; to be completed in a later version.

Appendix B: Traceability

Not required for v1.0; to be completed in a later version.

Appendix C: ML Requirement Mapping

Topic	Project use
Linear / Multiple Linear Regression	Resolution Days regression baselines.
Non-linear Regression	Compare non-linear relationships against linear models.
Gradient Descent	Optimization demonstration for regression training.
Regularization	Ridge/Lasso comparison for overfitting control.
Logistic Regression	High-delay vs normal-delay classification.
SVM	Alternative high-delay classifier.
Decision Trees	Interpretable prediction/classification.
Bagging	Random Forest/bagging comparison.
Boosting / AdaBoost	Sequential ensemble and AdaBoost classification experiments.
Naive Bayes + NLP	Problem Detail text classification using spaCy/NLTK preprocessing.
Sentiment Analysis	Complaint text polarity analysis where meaningful.
PCA + SVD	Dimensionality reduction and matrix-decomposition comparison.
K-Means + Hierarchical Clustering	Discovery of similar service-request groups.

CNN	Introductory experimental text-classification comparison if justified.
Cross-Validation	Reliable validation and model comparison.
Bias-Variance	Underfitting/overfitting analysis.
Confusion Matrix + Precision-Recall	Classification evaluation.

Current implementation position

The notebook has loaded a 10,000-row sample, selected 18 project-relevant columns, created and cleaned Resolution Days, and retained 3,901 rows with a usable target for supervised modeling. The next implementation stage is feature preparation/encoding, followed by baseline modeling and the requested ML experiments.